

Lesson 15: MLR Model Diagnostics

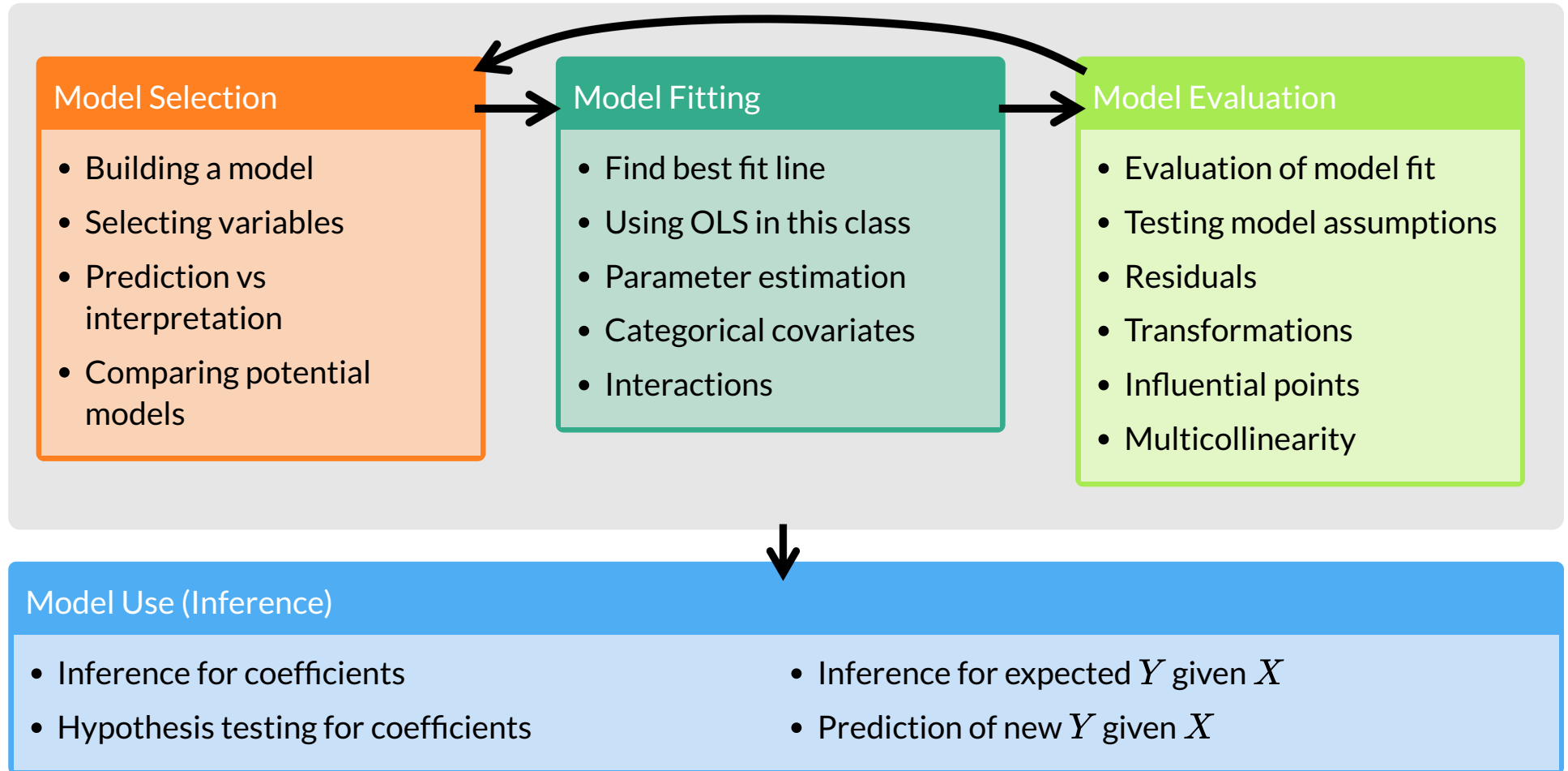
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Learning Objectives

1. Apply tools from SLR (Lesson 6: SLR Diagnostics) in MLR to evaluate LINE assumptions, including residual plots and QQ-plots
2. Apply tools involving standardized residuals, leverage, and Cook's distance from SLR (Lesson 7: SLR Diagnostics 2) in MLR to flag potentially influential points
3. Use Variance Inflation Factor (VIF) and its general form to detect and correct multicollinearity

Regression analysis process



Let's remind ourselves of the final model

- Our **final model** contains
 - Cell phones per 100 people
 - Basic sanitation (quartiles)
 - Freedom status
 - Income level
 - Vaccination rate
 - Happiness score
 - No interactions

► Display regression table for final model

Characteristic	Beta	95% CI	p-value
Cell phones per 100 people	0.01	-0.02, 0.04	0.7
Basic sanitation (%)			
[21.6,61.1]	—	—	
(61.1,92.8]	4.5	2.0, 7.0	<0.001
(92.8,98]	7.0	4.2, 9.8	<0.001
(98,100]	9.0	5.8, 12	<0.001
Freedom status			
NF	—	—	
PF	1.2	-0.72, 3.2	0.2
F	-0.35	-2.9, 2.2	0.8
Income level			
Low income	—	—	
Lower middle income	-0.25	-3.4, 2.9	0.9
Upper middle income	-0.11	-4.2, 4.0	>0.9
High income	3.5	-1.8, 8.8	0.2
Vaccination rate (%)	0.03	-0.08, 0.14	0.6
Happiness score	0.14	0.03, 0.25	0.014
Abbreviation: CI = Confidence Interval			

It's a lot to visualize

- Part of the reason why we discussed model diagnostics in SLR was so that we could have accompanying visuals to help us understand
- With 6 variables in our final model, it is hard to scatterplots to visualize linearity, outliers, and influential points
- I highly encourage you revisit the SLR lessons ([SLR: Checking model assumptions](#) and [SLR: Diagnostics](#) to help understand these notes

Remember our friend `augment()`?

- Run `final_model` through `augment()` (`final_model` is input)
 - So we assigned `final_model` as the output of the `lm()` function
- Will give us values about each observation in the context of the fitted regression model
 - cook's distance (`.cooks`), $\hat{Y}_i$ (`.fitted`), leverage (`.hat`), residuals (`.resid`), std residuals (`.std.resid`)

```
1 aug = augment(final_model)
2 head(aug) %>% relocate(.fitted, .resid, .std.resid, .hat, .cooks, .after = life_exp)
```

A tibble: 6 × 13

	life_exp	.fitted	.resid	.std.resid	.hat	.cooks	cell_phones_100	BS_q
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>
1	62.6	62.8	-0.166	-0.0506	0.342	0.000111	56.3	(61.1,92...
2	76.1	75.1	0.954	0.251	0.110	0.000647	98.4	(98,100]
3	73.4	80.3	-6.88	-1.87	0.166	0.0581	196.	(98,100]
4	75.4	75.1	0.267	0.0700	0.109	0.0000499	131.	(98,100]
5	73.7	75.5	-1.88	-0.486	0.0829	0.00178	131.	(98,100]
6	71.4	71.0	0.374	0.0981	0.108	0.0000971	108.	(92.8,98]

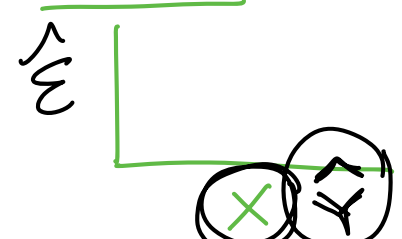
i 5 more variables: freedom_status <fct>, income_level_4 <fct>,
vax_rate <dbl>, happiness_score <dbl>, .sigma <dbl>

RDokumentation on the `augment()` function.

Learning Objectives

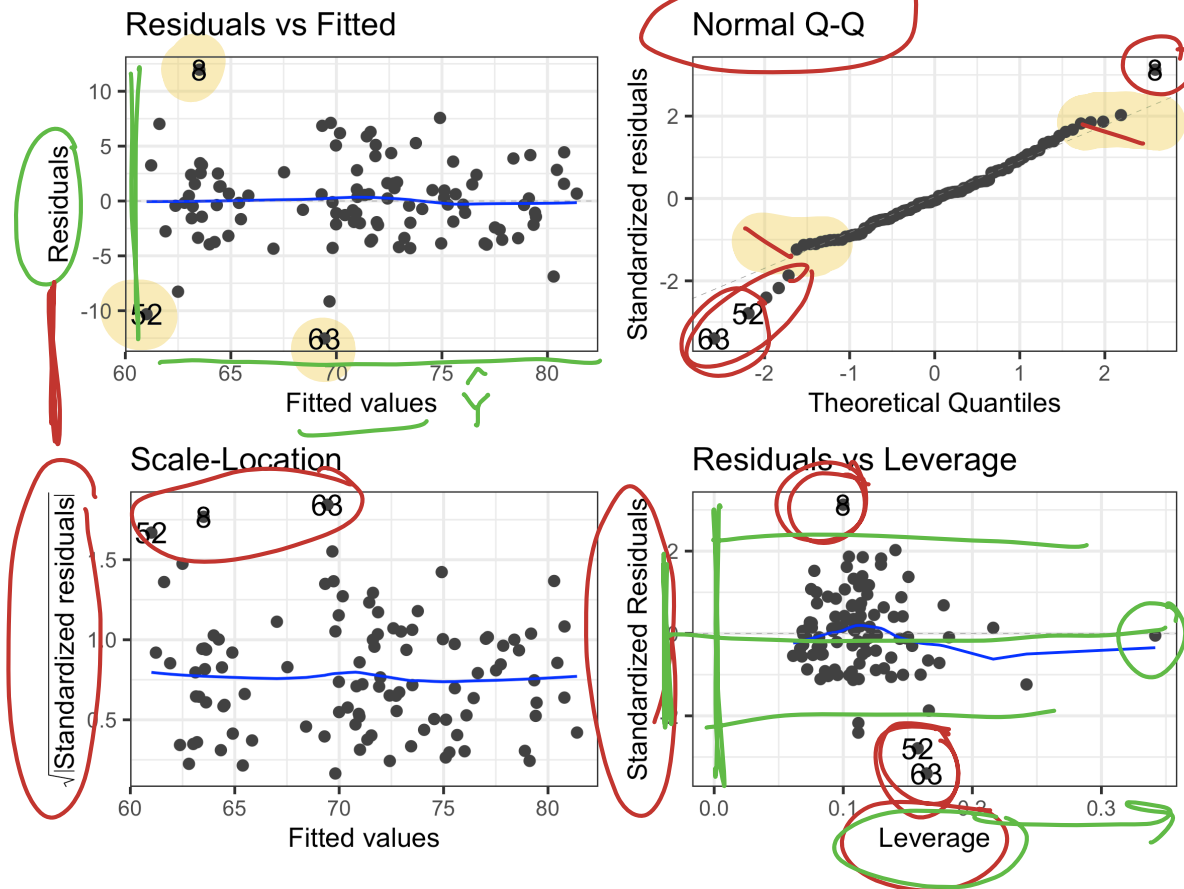
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Summary of the assumptions and their diagnostic tool

Assumption	What needs to hold?	Diagnostic tool
Linearity	Relationship between each X and Y is linear	<u>Scatterplot of Y vs. X</u>
Independence	Observations are independent from each other	Study design ✓
Normality	Residuals (and thus $Y X_1, X_2, \dots, X_p$) are normally distributed	<u>QQ plot of residuals</u> <u>Distribution of residuals</u>
<u>Equality of variance</u>	Variance of residuals (and thus $Y X_1, X_2, \dots, X_p$) is same across fitted values (homoscedasticity)	Residual plot 

autoplot() to examine equality of variance and Normality

```
1 library(ggfortify)
2 autoplot(final_model) + theme(text=element_text(size=20))
```



autoplot() to examine equality of variance and Normality

```
1 library(ggfortify)
2 autoplot(final_model) + theme(text=element_text(size=
```

Looks like 2 obs are flagged:

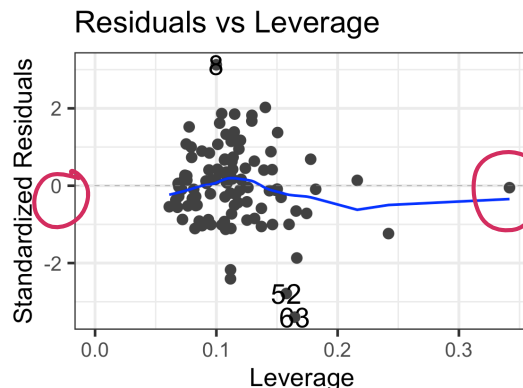
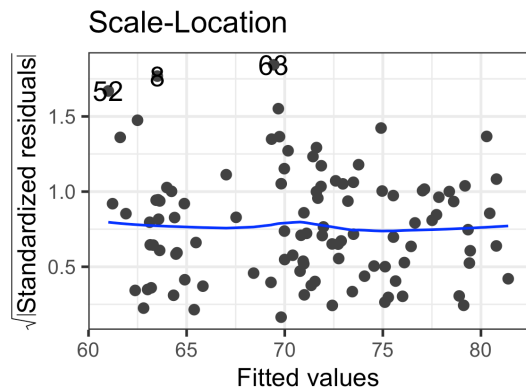
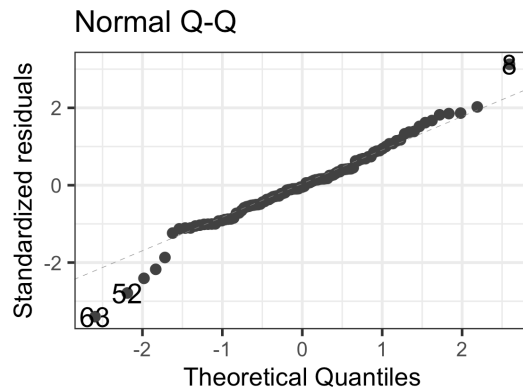
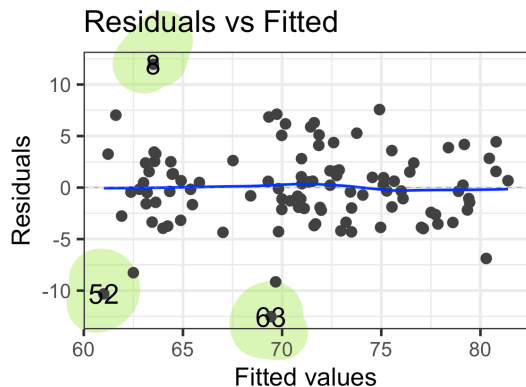
- 8: Bangladesh
- 52: Lesotho
- 66: Malawi

Without them, QQ-plot and residual plot look good

- Points on QQ-plot are close to identity line
- Residuals have pretty consistent spread across fitted values

But don't take them out!!!

- Instead, discuss what may be missing in our regression model that is not capturing the characteristics of these countries



Poll Everywhere Question 1

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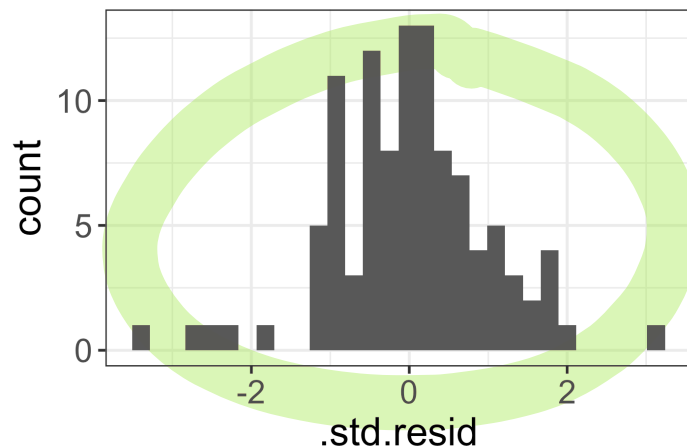
Identifying outliers

Internally standardized residual

$$r_i = \frac{\hat{\epsilon}_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}}$$

- We flag an observation if the standardized residual is “large”
 - Different sources will define “large” differently
 - PennState site uses $|r_i| > 3$
 - `autoplot()` shows the 3 observations with the highest standardized residuals
 - Other sources use $|r_i| > 2$, which is a little more conservative

```
1 ggplot(data = aug) +  
2   geom_histogram(aes(x = .std.resid))
```



Countries that are outliers ($|r_i| > 3$)

- We can identify the countries that are outliers

```
1 aug %>% relocate(.std.resid, .after = territory) %>%  
2   filter(abs(.std.resid) > 3) %>% arrange(desc(abs(.std.resid)))
```

A tibble: 2 × 14

	territory <chr>	.std.resid <dbl>	life_exp <dbl>	cell_phones_100 <dbl>	BS_q <dbl>	freedom_status <fct>
1	Mozambique	-3.40	56.9	45.8	(61.1,92.8]	PF
2	Bangladesh	3.13	75.5	110.	[21.6,61.1]	PF

i 8 more variables: income_level_4 <fct>, vax_rate <dbl>,
happiness_score <dbl>, .fitted <dbl>, .resid <dbl>, .hat <dbl>,
.sigma <dbl>, .cooksdi <dbl>

Leverage h_i

- Values of leverage are: $0 \leq h_i \leq 1$
- We flag an observation if the leverage is “high”
 - **Only good for SLR:** Some textbooks use $h_i > 4/n$ where n = sample size
 - **Only good for SLR:** Some people suggest $h_i > 6/n$
 - **Works for MLR:** $h_i > 3p/n$ where p = number of regression coefficients

```
1 aug = aug %>% relocate(.hat, .after = cell_phones_100)
2 aug %>% arrange(desc(.hat)) %>% head(5)
```

```
# A tibble: 5 × 14
```

	territory	life_exp	cell_phones_100	.hat	BS_q	freedom_status	income_level_4
	<chr>	<dbl>	<dbl>	<dbl>	<fct>	<fct>	<fct>
1	Afghanistan	62.6	56.3	0.342	(61.... NF		Low income
2	Botswana	62.7	178.	0.242	(61.... F		Upper middle ...
3	Yemen	66.3	46.5	0.216	(61.... NF		Low income
4	Montenegro	75.7	207.	0.182	(98,... PF		Upper middle ...
5	Gabon	66.9	123.	0.178	[21.... NF		Upper middle ...

```
# i 7 more variables: vax_rate <dbl>, happiness_score <dbl>, .fitted <dbl>,
# .resid <dbl>, .sigma <dbl>, .cooks_d <dbl>, .std.resid <dbl>
```

Countries with high leverage ($h_i > 3p/n$)

- We can look at the countries that have high leverage: there are NONE

```
1 n = nrow(gapm2); p = length(final_model$coefficients) - 1
2 aug %>%
3   filter(.hat > 3*p/n) %>%
4   arrange(desc(.hat))
```

A tibble: 1 × 14

territory	life_exp	cell_phones_100	.hat	BS_q	freedom_status	income_level_4
<chr>	<dbl>	<dbl>	<dbl>	<fct>	<fct>	<fct>
1 Afghanistan	62.6	56.3	0.342	(61.... NF		Low income

i 7 more variables: vax_rate <dbl>, happiness_score <dbl>, .fitted <dbl>,
.resid <dbl>, .sigma <dbl>, .cooks_d <dbl>, .std.resid <dbl>

Cook's distance

- Measures the overall influence of an observation
- Attempts to measure how much **influence a single observation has** over the fitted model
 - Measures **how coefficient estimates change** when the i^{th} observation is removed from the model
 - Combines leverage and outlier information

The Cook's distance for the i^{th} observation

$$d_i = \frac{h_i}{2(1 - h_i)} \cdot r_i^2$$

where h_i is the leverage and r_i is the studentized residual

- Another rule for Cook's distance that is not strict:
 - Investigate observations that have $d_i > 1$
- Cook's distance values are already in the augment tibble: `.cooksd`

Identifying points with high Cook's distance

- No countries with high Cook's distance

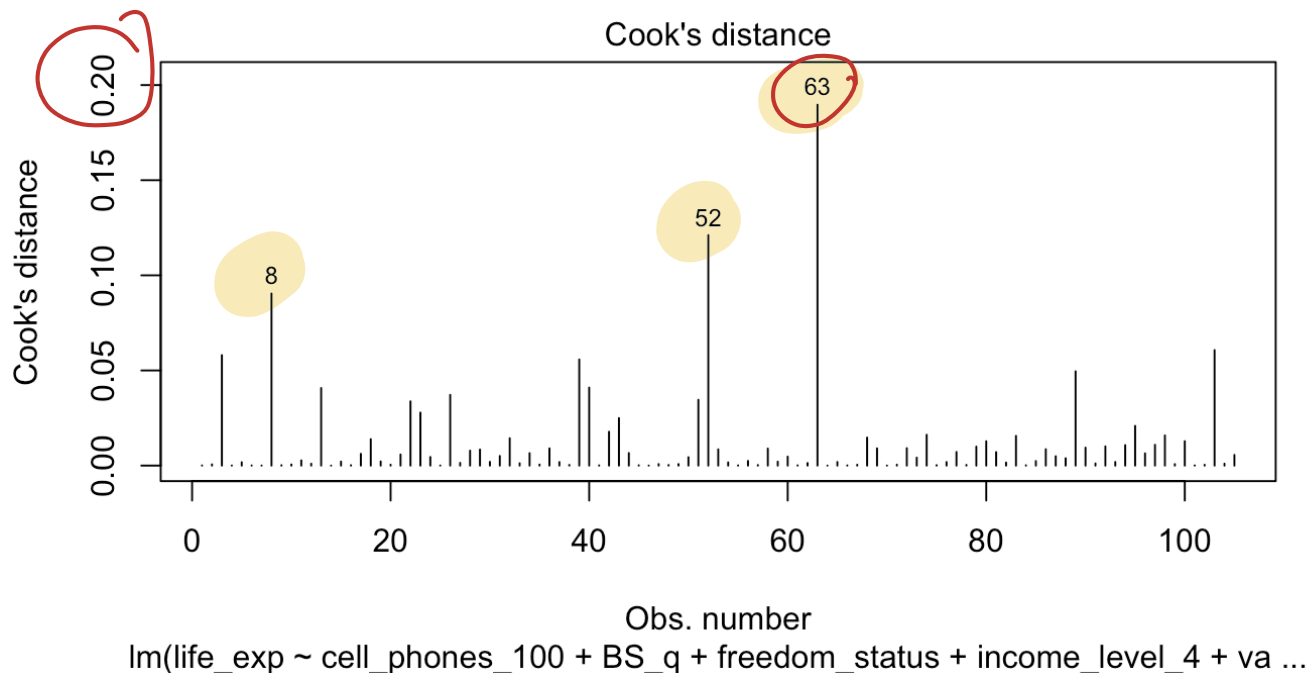
```
1 aug = aug %>% relocate(.cooksd, .after = territory)
2 aug %>% arrange(desc(.cooksd)) %>% filter(.cooksd > 1)
```

```
# A tibble: 0 × 14
```

```
# i 14 variables: territory <chr>, .cooksd <dbl>, life_exp <dbl>,  
# cell_phones_100 <dbl>, .hat <dbl>, BS_q <fct>, freedom_status <fct>,  
# income_level_4 <fct>, vax_rate <dbl>, happiness_score <dbl>, .fitted <dbl>,  
# .resid <dbl>, .sigma <dbl>, .std.resid <dbl>
```

Plotting Cook's Distance

```
1 # plot(model) shows figures similar to autoplot() but adds Cook's distance
2 plot(final_model, which = 4)
```



- Identify 3 highest Cook's distance: 8, 52, ~~66~~ (Bangladesh, Lesotho, ~~Malawi~~)
63

How do we deal with influential points?

- If an observation is influential, we can **check data errors**:
 - Was there a data entry or collection problem?
 - If you have reason to believe that the observation does not hold within the population (or gives you cause to redefine your population)
- If an observation is influential, we can **check our model**:
 - Did you leave out any important predictors?
 - Should you consider adding some interaction terms?
 - Is there any nonlinearity that needs to be modeled?
- Basically, deleting an observation should be justified outside of the numbers!
 - If it's an honest data point, then it's giving us important information!
- **Means we will need to discuss the limitations of our model**
 - For example: Think about measurements that might help explain life expectancy that are NOT in our model
- A really well thought out explanation from StackExchange

Poll Everywhere Question 2

When we have detected problems in our model...

- We have talked about influential points
- We have talked about identifying issues with our LINE assumptions

What are our options once we have identified issues in our linear regression model?

- Are we missing a crucial measure in our dataset?
- Try categorization or transformation (or numeric variables) if there is an issue with linearity or normality
 - Addressed in model selection
- Try a weighted least squares approach if unequal variance (oof, not enough time for us to get to)
- Try a robust estimation procedure if we have a lot of outlier issues (outside scope of class)

↳ robust SE's

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What is multicollinearity? (adapted from parts of STAT 501 page)

So far, we've been ignoring something very important: multicollinearity

Multicollinearity

Two or more covariates in a multivariable regression model are highly correlated

- Types of multicollinearity

- **Structural multicollinearity**

- Mathematical artifact caused by creating new covariates from other covariates
 - For example: If we have age, and decide to transform age to include age-squared
 - Then we have age and age-squared in the model: age-squared is perfectly predicted by age!

} center

- **Data-based multicollinearity**

- Result of a poorly designed experiment, reliance on purely observational data, or the inability to manipulate the system on which the data are collected
 - Or we are using two variables that are practically measuring the same thing

Poll Everywhere Question 3

Why is multicollinearity a problem?

In linear regression, **depending on the other predictors in the model**, the following will change:

- Estimated regression coefficient of any one variable
 - Not necessarily bad, but a big change might be an issue
 - Hypothesis tests for any coefficient may yield different conclusions
 - Contribution of any one predictor variable in reducing sum of squared errors
- 

When there is multicollinearity in our model:

- **Precision of the estimated regression coefficients or correlated covariates decreases a lot**
 - Basically, **standard error increases and confidence intervals get wider**, which means we're not as confident in our estimate anymore
 - Because highly correlated covariates are **not adding much more information**, but are constraining our model more

Did you notice anything about all the consequences of multicollinearity?

- All consequences relate to estimating a regression coefficient **precisely**
 - Recall that precision is linked to analysis **goals of association and interpretability**
 - See lesson on **Model Selection**
- Multicollinearity is *not really an issue* when our **goal is prediction**
 - Highly correlated covariates/predictors will not hurt our prediction of an outcome

How do we detect multicollinearity?

- **Variance inflation factors (VIF):** quantifies how much the variance of the estimated coefficient for covariate k increases
 - Increases: from SLR with only covariate k to MLR with all other covariates
- General rule of thumb
 - $VIF < 4$: Good!
 - $4 < VIF < 10$: Warrent investigation (but most people aren't investigating this...)
 - $VIF > 10$: Requires correction
 - Influencing regression coefficient estimates

VIF

$$VIF = \frac{1}{1 - R_k^2}$$

R_k^2 is the R^2 -value obtained by regressing the k^{th} covariate/predictor on the remaining predictors

Let's apply it to our final model

- Naive way to calculate this in R:

```
1 library(rms)
2 rms::vif(final_model)
```

cell_phones_100	BS_q(61.1,92.8]
1.903507	1.920420
BS_q(92.8,98]	BS_q(98,100]
2.428357	3.171019
freedom_statusPF	freedom_statusF
1.517866	2.081666
income_level_4Lower middle income	income_level_4Upper middle income
3.672232	5.994002
income_level_4High income	vax_rate
6.812730	1.428216
happiness_score	
2.765792	

- All $VIF < 10$
- Problem: multi-level covariates (CO2 Emissions and income level) have different VIF's even though they should be considered one variable

Let's apply it to our final model *correctly* (1/2)

- Calculate the GVIF and, more importantly, the $GVIF^{1/(2 \cdot df)}$
- GVIF is the R^2 -value for regressing a covariate's group indicators on the remaining covariates
 - Captures the correlation between covariates better
- $GVIF^{1/(2 \cdot df)}$ helps standardize GVIF based on how many levels each categorical covariate has
 - I'll refer to this as df-corrected GVIF or standardized GVIF
 - If continuous covariate, $GVIF^{1/(2 \cdot df)} = \sqrt{GVIF}$

```
1 library(car)
2 car::vif(final_model)
```

	GVIF	Df	GVIF^(1/(2*Df))
cell_phones_100	1.903507	1	1.379676
BS_q	2.658818	3	1.177013
freedom_status	1.660933	2	1.135241
income_level_4	5.648213	3	1.334500
vax_rate	1.428216	1	1.195080
happiness_score	2.765792	1	1.663067

Let's apply it to our final model *correctly* (2/2)

- If continuous covariate, $GVIF^{1/(2 \cdot df)} = \sqrt{GVIF}$
- So we can square $GVIF^{1/(2 \cdot df)}$ and set VIF rules
- OR: we can correct any $GVIF^{1/(2 \cdot df)} > \sqrt{10} = 3.162$

```
1 car::vif(final_model)
```

	GVIF	Df	GVIF^(1/(2*Df))
cell_phones_100	1.903507	1	1.379676
BS_q	2.658818	3	1.177013
freedom_status	1.660933	2	1.135241
income_level_4	5.648213	3	1.334500
vax_rate	1.428216	1	1.195080
happiness_score	2.765792	1	1.663067

- All of these covariates are okay! No multicollinearity to correct in this dataset!

But what if we do need to make corrections for multicollinearity?

- We have been dealing with **data-based multicollinearity** in our example
- If we had issues with multicollinearity, then what are our options?
 - Remove the variable(s) with large VIF
 - Use expert knowledge in the field to decide
- If one variable has a large VIF, then there is usually another one or more variables with large VIFs
 - Basically, all the covariates that are correlated will have large VIFs
- Example: our two largest GVIFs were for world region and income levels
 - Hypothetical: their $GVIF^{1/(2 \cdot df)} > 3.162$
 - Remove one of them
 - I'm no expert, but from more of a data equity lens, there's a lot of generalizations made about world regions
 - I think relying on the income level of a country might give us more information as well

What about structural multicollinearity?

- **Structural multicollinearity**
 - Mathematical artifact caused by creating new covariates from other covariates
- For example: If we have age, and decide to transform age to include age-squared
 - Then we have age and age-squared in the model: age-squared is perfectly predicted by age!
 - By having the untransformed and transformed covariate in the model, they are inherently correlated!
- **Best practice to reduce the correlation: center your covariate**
 - By centering age, we no longer have a one-to-one connection between age and age-squared
 - If centered at 40yo: a 35yo has centered age of -5 and a 45yo centered age of 5, but both have age-squared of 25
- **Check out the Penn State site** for a work through of an example with VIFs

Summary of multicollinearity

- Correlated covariates/predictors will hurt our model's precision and interpretations of coefficients
- We need to check for multicollinearity by using VIFs or GVIFs
- If $VIF > 10$ or $GVIF^{1/(2 \cdot df)} > 3.162$, we need to do something about the covariates
 - Data based: remove one the of correlated variables
 - Structural based: centering usually fixes it

Regression analysis process

